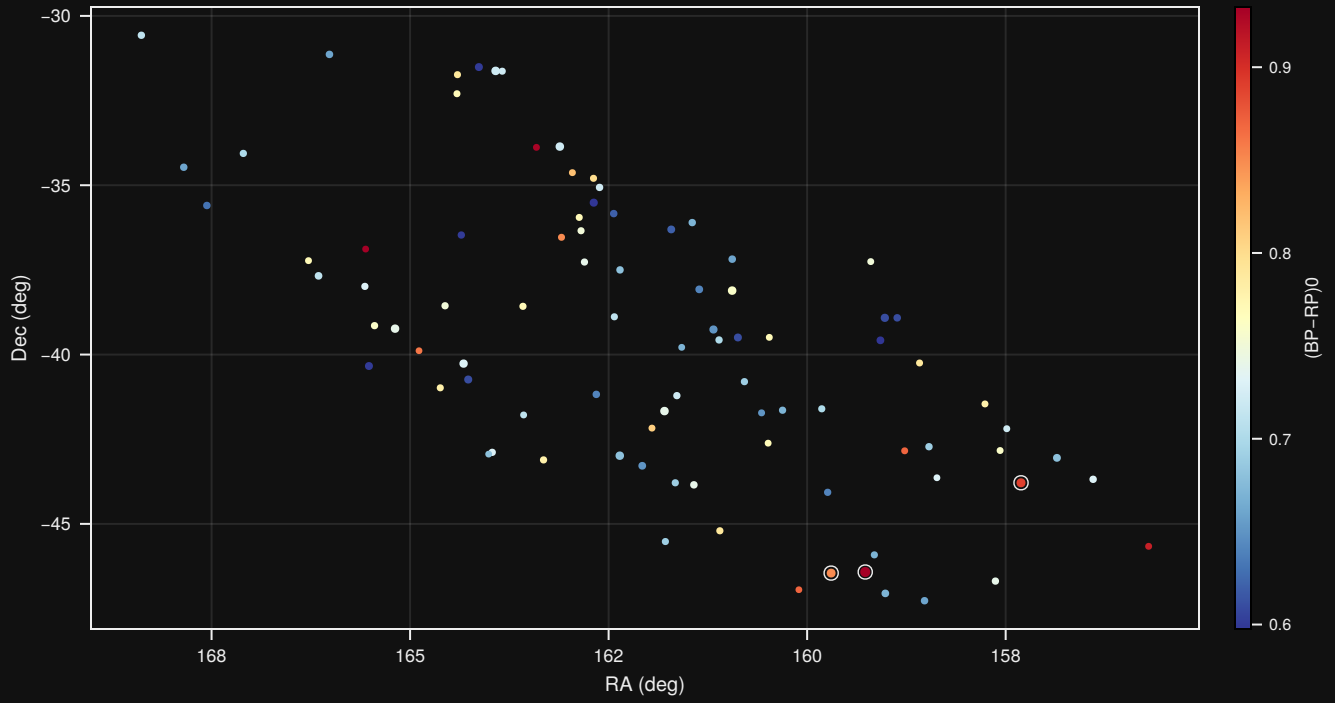
# New-12

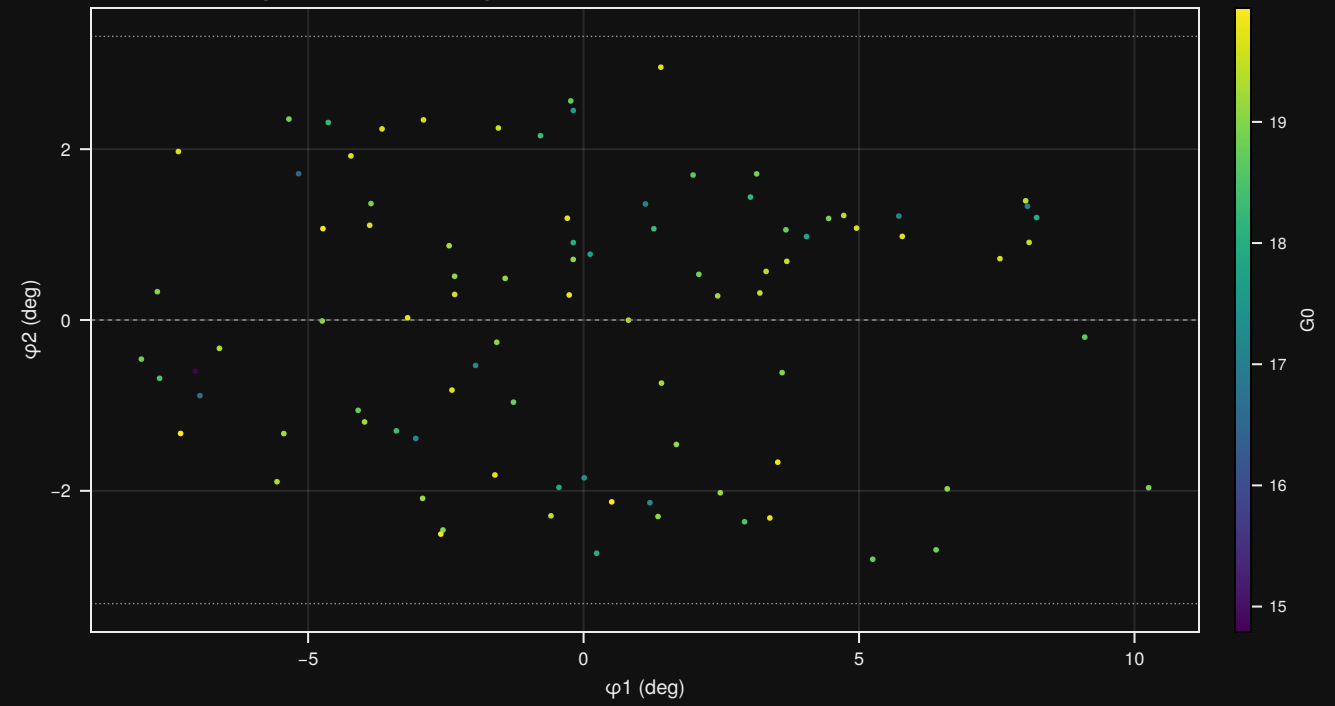


| | |
|---|---|
| **MEMBERSHIP** | |
| Stars in catalog | 89 |
| with Gaia GSP-Phot | 29 (33%) |
| with radial velocity | 3 |
| Stream id (sID) | 37 |
| **GEOMETRY · great-circle fit** | |
| Centroid RA, Dec | 162.0, -39.6 deg |
| Centroid l, b | 278.5, 17.0 deg |
| Pole RA, Dec | 85.6, 15.8 deg |
| Length (φ1, 2–98%) | 16.1 deg |
| Width σ(φ2), robust | 1.66 deg |
| **KINEMATICS** | |
| pmRA* med ± σ | 1.22 ± 0.86 mas/yr |
| pmDE med ± σ | 6.69 ± 1.40 mas/yr |
| μφ1* med ± σ | 6.70 ± 1.37 mas/yr |
| μφ2 med ± σ | 1.18 ± 0.49 mas/yr |
| Parallax (cat.) med | 0.148 ± 0.279 mas |
| Parallax (wtd, ZP-corr.) | 0.177 ± 0.018 mas |
| Distance | d = 5.6 ± 0.6 kpc |
| Physical length | 1.59 kpc |
| RV median ± σ(MAD) | 299.6 ± 2.0 km/s |
| RV gradient dV/dφ1 | — |
| **PHOTOMETRY · POPULATIONS** | |
| G0 range (median) | 14.8–19.9 (19.1) |
| (BP−RP)0 median | 0.72 |
| GSP-Phot [M/H] (n=29) | -1.82 ± 0.57 |
| GSP-Phot Teff median | 5923 K |
| RUWE > 1.4 | 0.0 % |
| PM clusters (largest) | 5 (28 stars) |
| **IBATA+2024 · TABLE 3** | |
| n catalog \| paper | 89 \| 89 |
| n_RV catalog \| paper | 3 \| 3 |
| <[Fe/H]> paper | -1.85 |
| Stellar mass (paper) | 580 Msun |
| Pole offset vs paper | 2.0 deg |

**Sky map · catalog members (n = 89) · size ∝ brightness · ring = has RV**

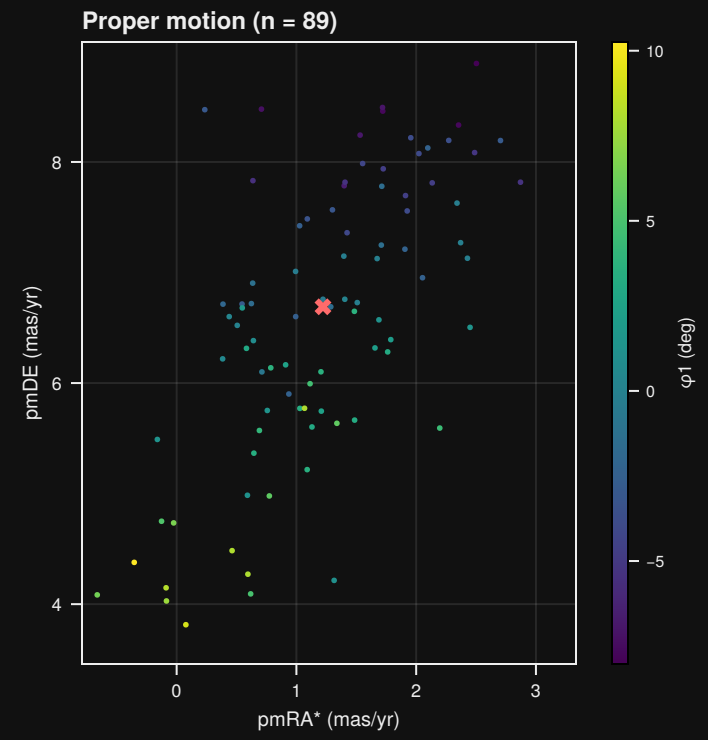


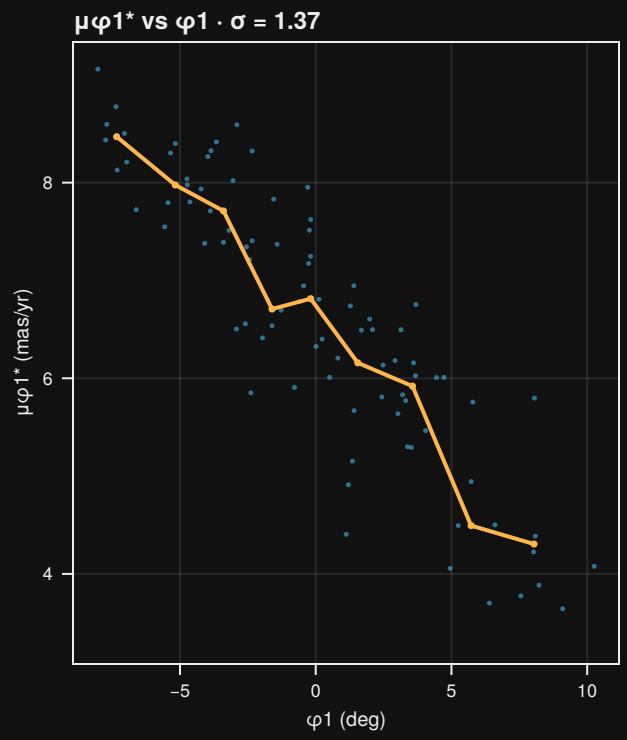

**Stream frame (great-circle fit) · length 16.1° · width σ 1.66°**

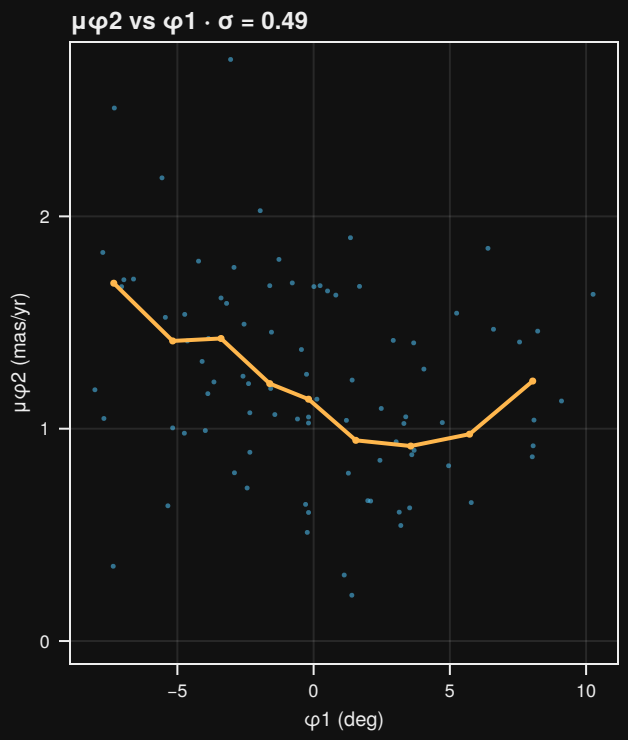


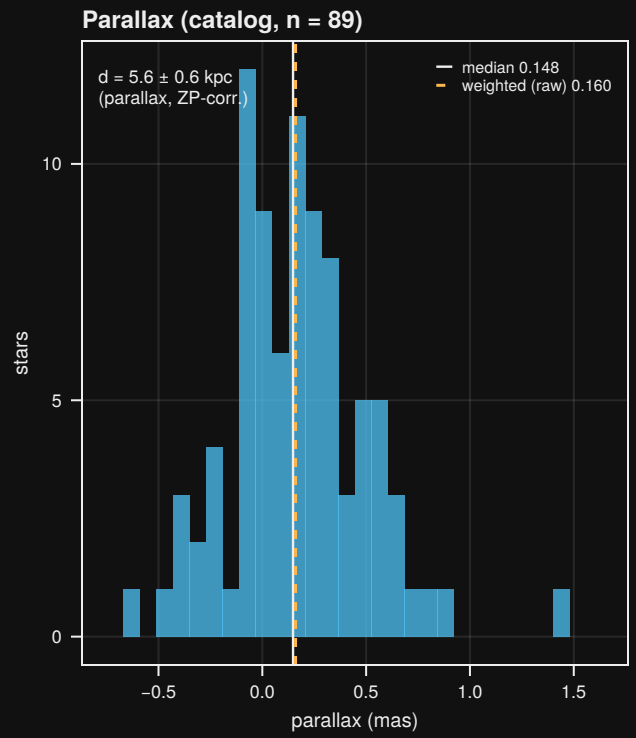
# New-12

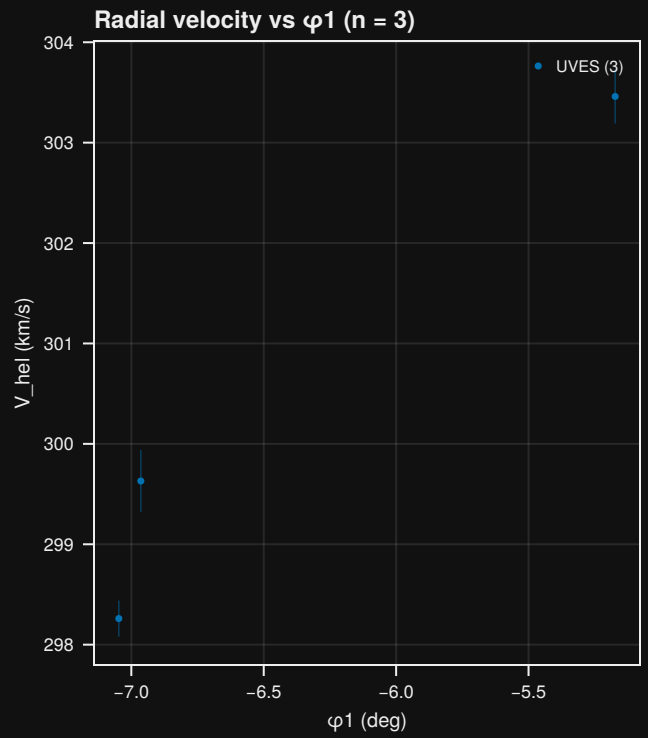
Kinematics · proper motion · parallax · radial velocity

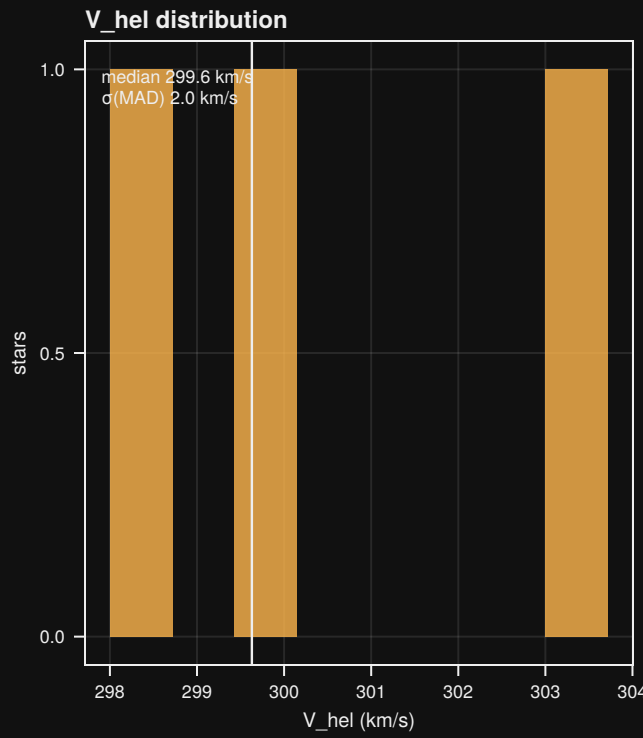


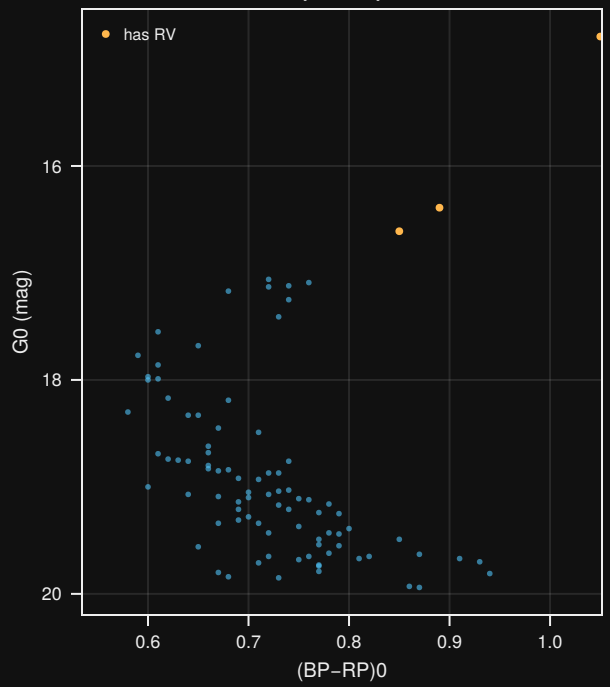



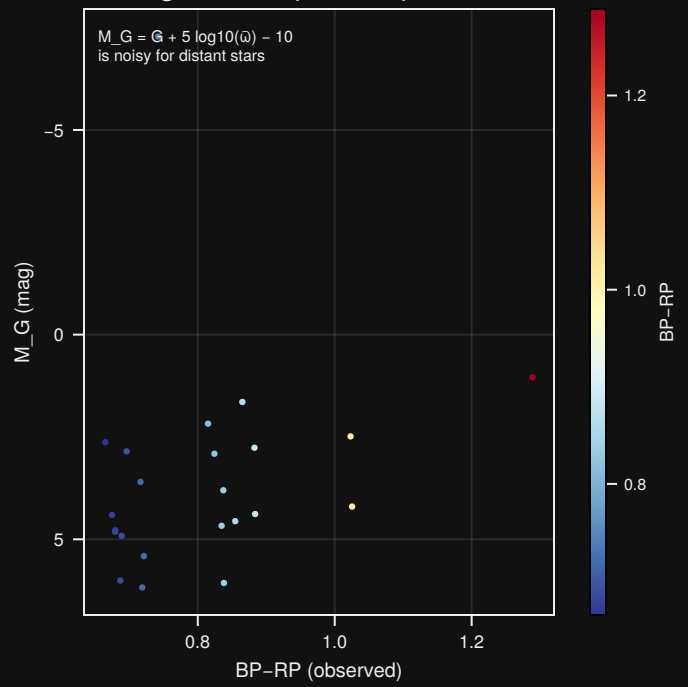
# New-12



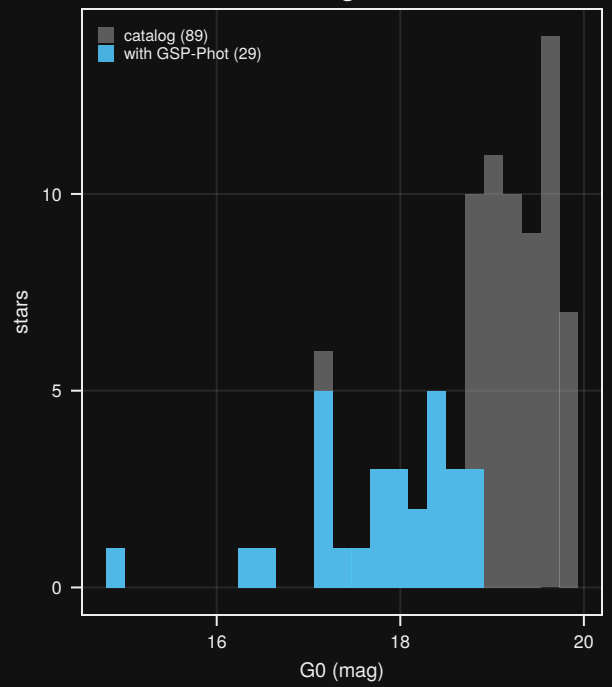
## CMD · dereddened (n = 89)



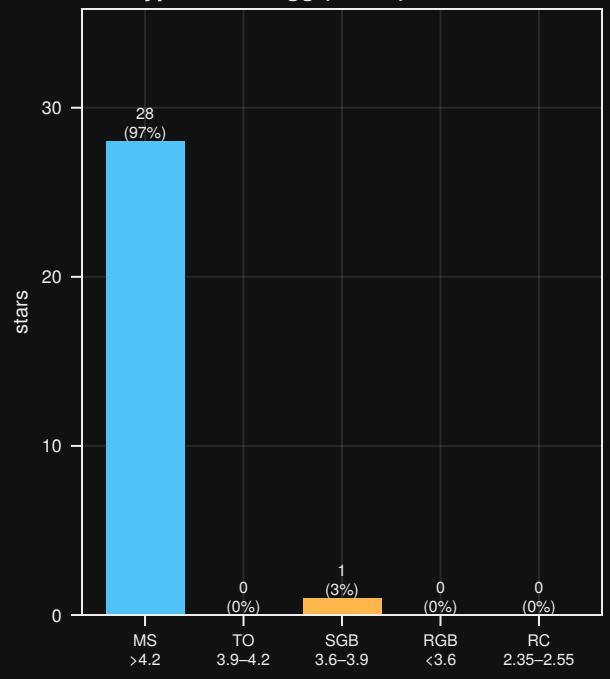

## HR diagram · ϖ>0 (n = 23/29)



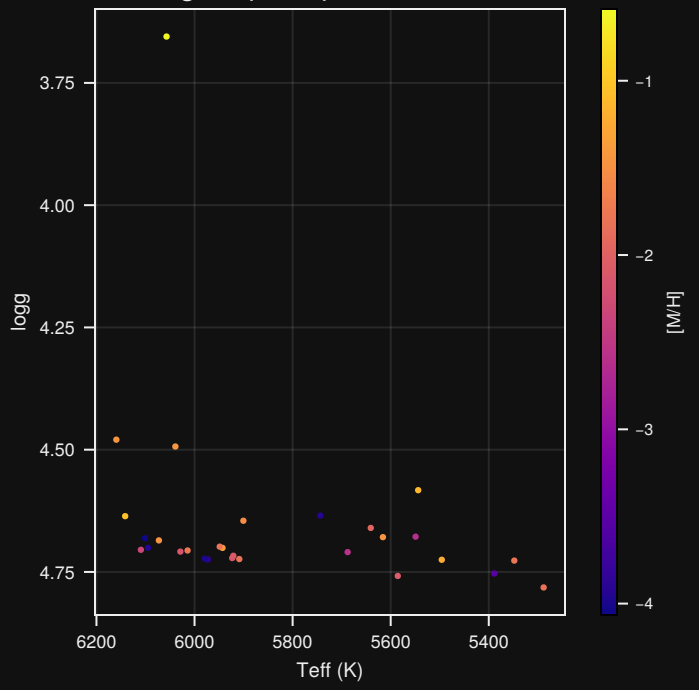

## G0 & GSP-Phot coverage



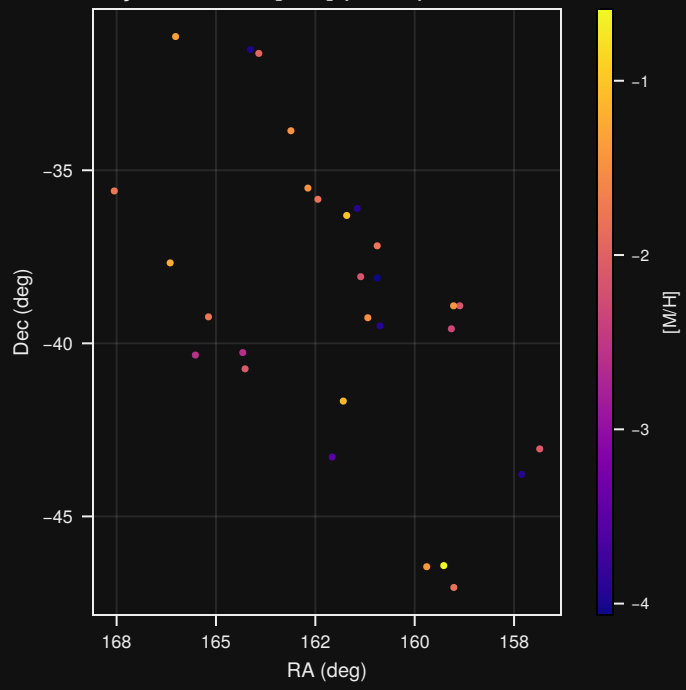

## Star types from logg (n = 29)




## Kiel diagram (n = 29)




## Sky · GSP-Phot [M/H] (n = 29)

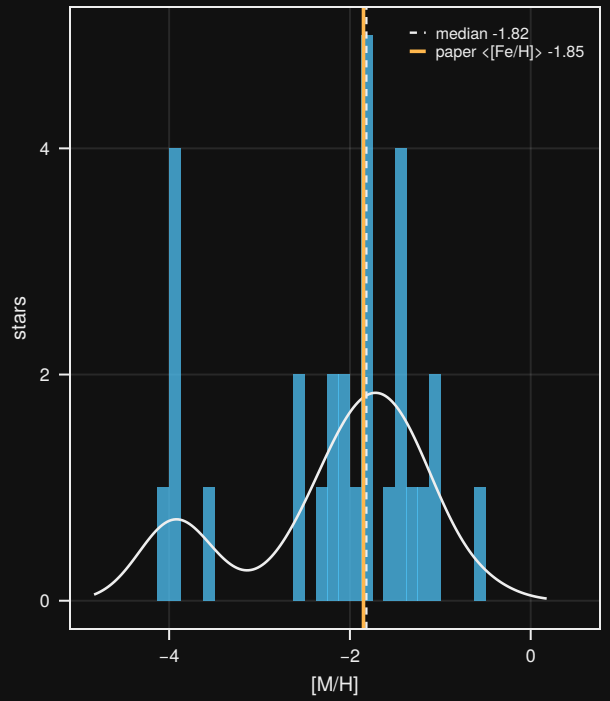



GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

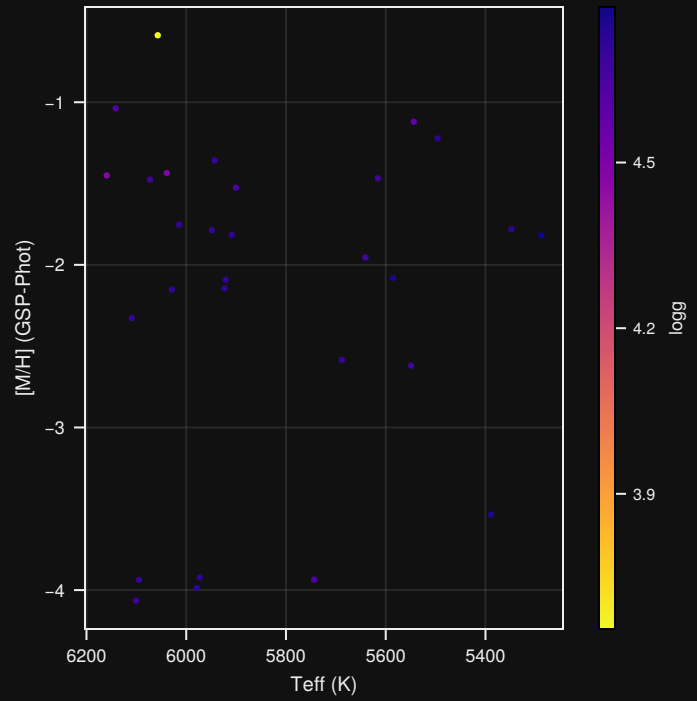
# New-12

Metallicity · age · astrometric quality

## [M/H] GSP-Phot (n = 29)




## Teff vs [M/H] (n = 29)



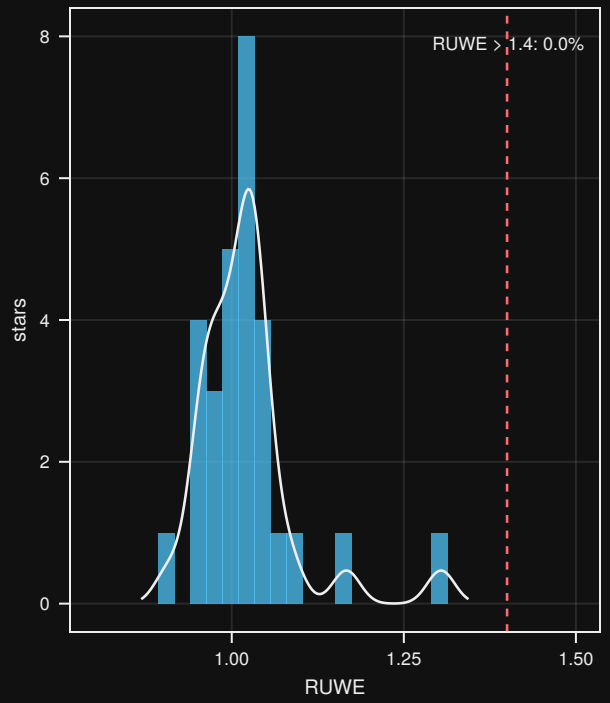

## FLAME age (n = 1)

FLAME ages for only 1 star(s)
(FLAME needs good parallaxes)

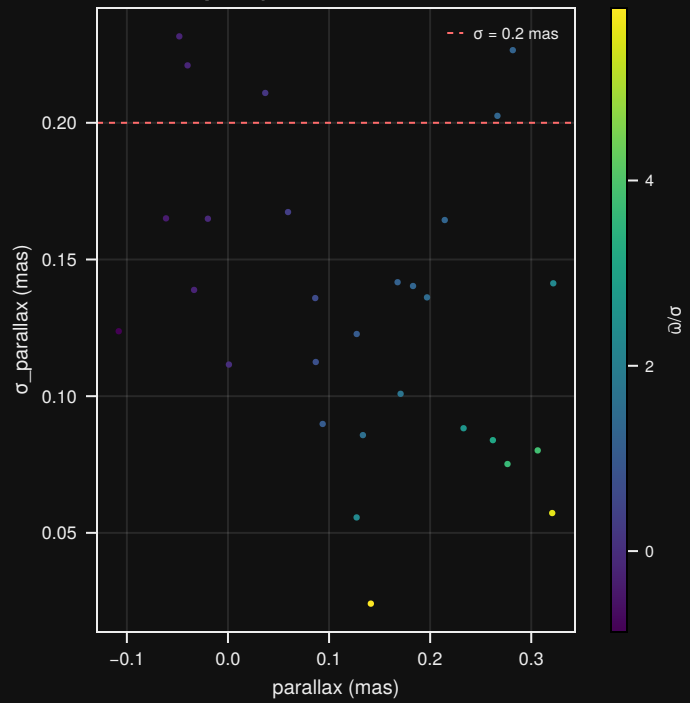
## Age–metallicity (n = 1)

n < 3

## RUWE (n = 29)



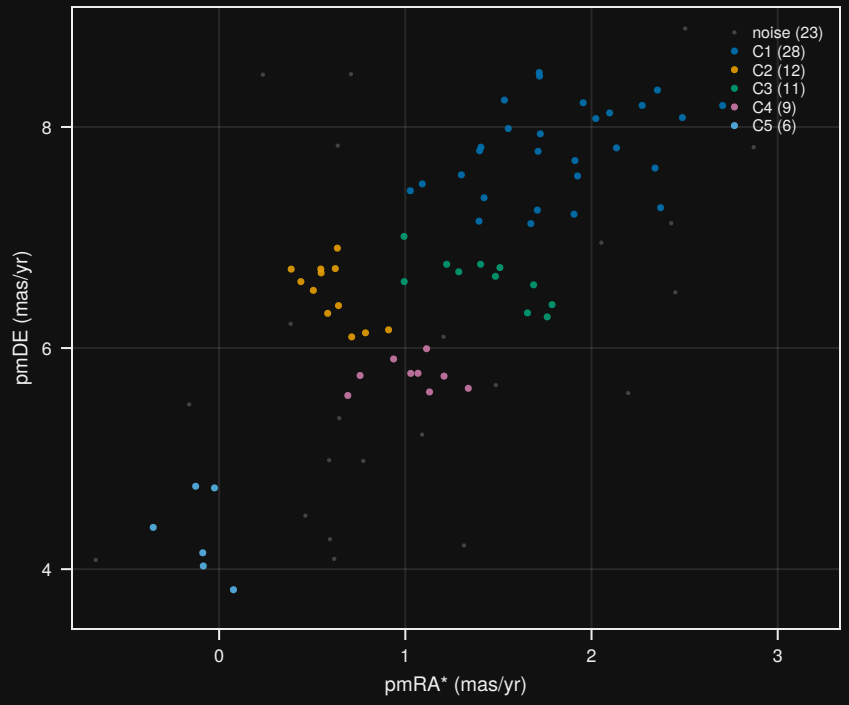

## Parallax quality (n = 29)



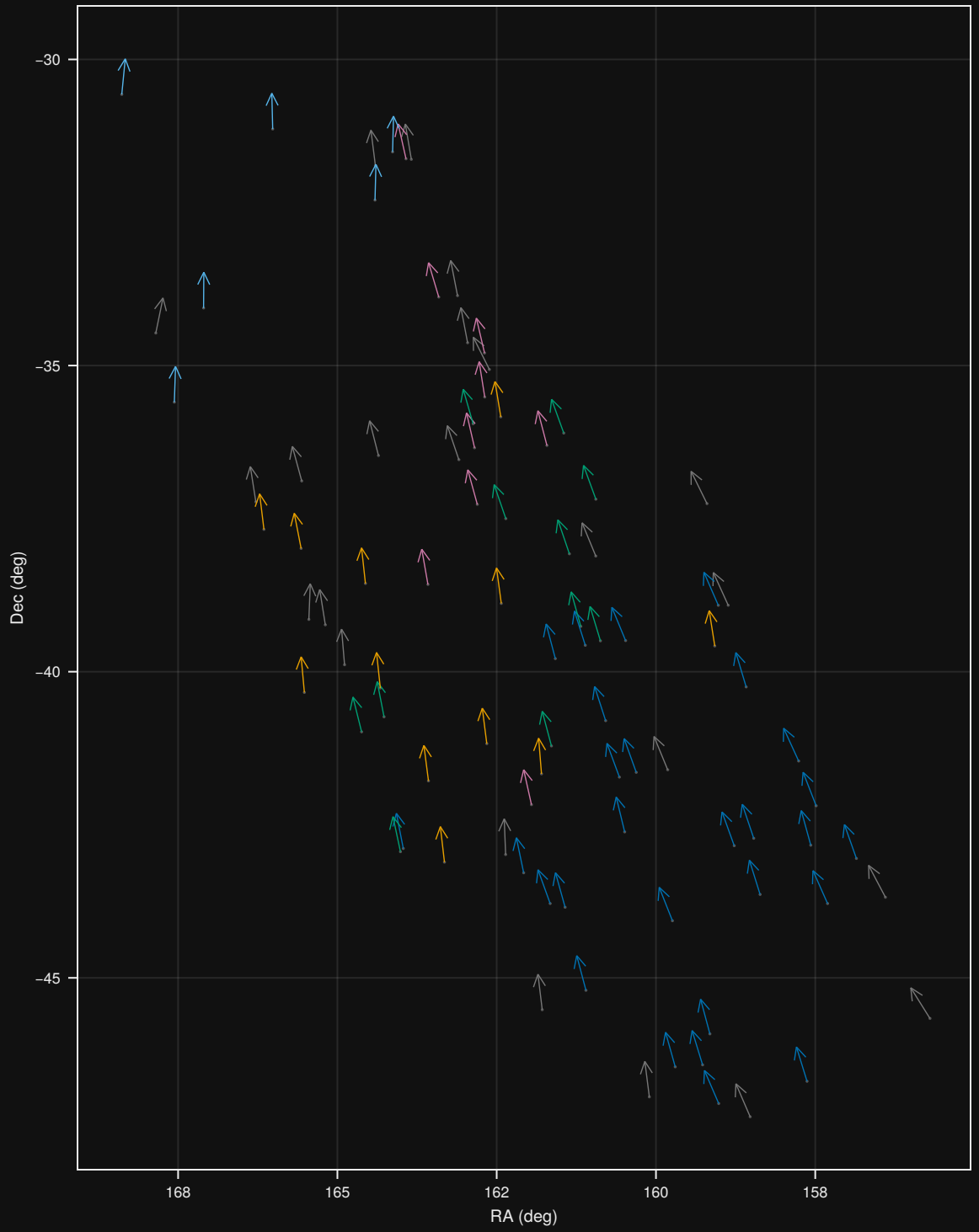
# New-12



**PM clustering · HDBSCAN · 5 cluster(s) · noise 26%**




**Proper-motion directions (arrows) coloured by cluster**



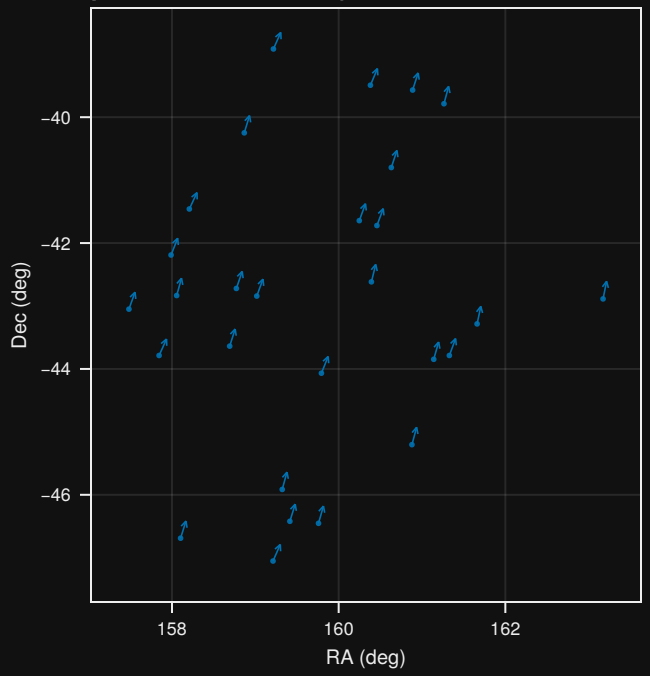

## Cluster summary

| id | n | pmRA | pmDE | σpmRA | [M/H] | n_RV |
|---|---|---|---|---|---|---|
| noise | 23 | 0.77 | 5.67 | 0.80 | -1.96 | 0 |
| C1 | 28 | 1.72 | 7.80 | 0.46 | -1.82 | 3 |
| C2 | 12 | 0.60 | 6.56 | 0.11 | -2.07 | 0 |
| C3 | 11 | 1.48 | 6.65 | 0.30 | -2.12 | 0 |
| C4 | 9 | 1.07 | 5.75 | 0.19 | -1.47 | 0 |
| C5 | 6 | -0.09 | 4.26 | 0.07 | -1.75 | 0 |

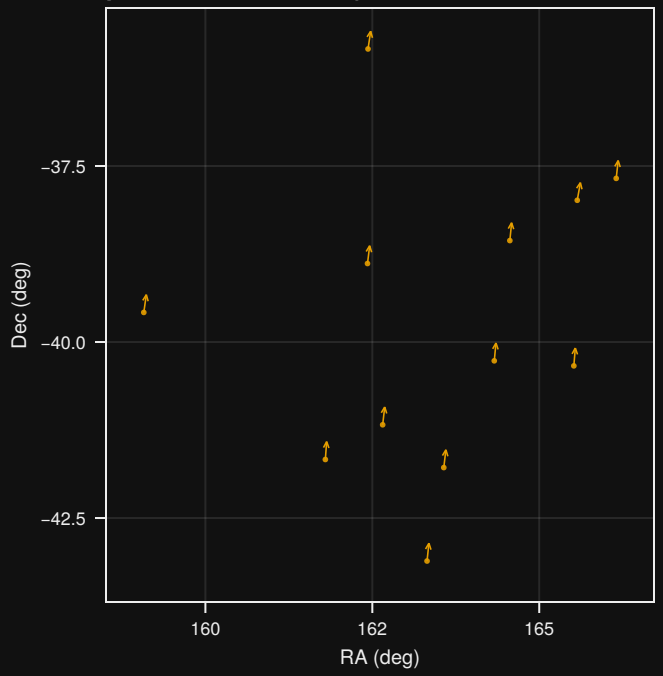
# New-12

Per-cluster proper-motion vector maps

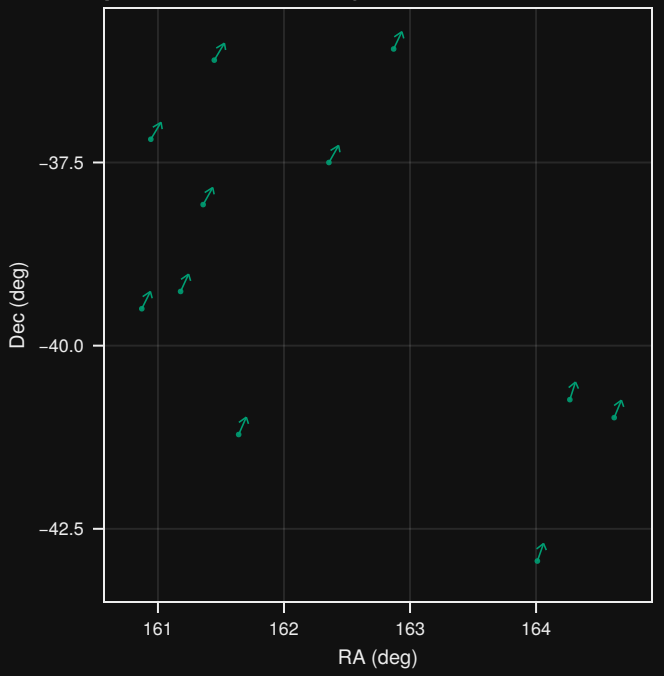
**Cluster C1 · n = 28**
**pm = (1.72, 7.80) mas/yr**



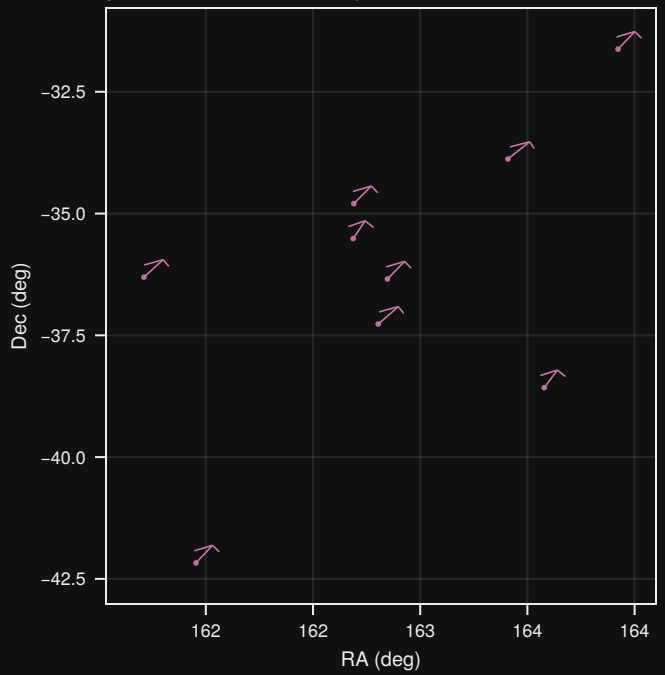

**Cluster C2 · n = 12**
**pm = (0.60, 6.56) mas/yr**



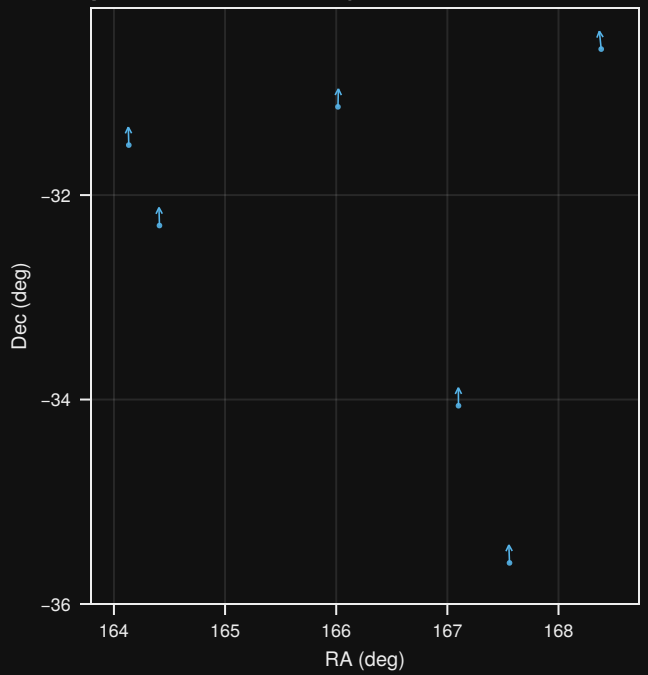

**Cluster C3 · n = 11**
**pm = (1.48, 6.65) mas/yr**




**Cluster C4 · n = 9**
**pm = (1.07, 5.75) mas/yr**




**Cluster C5 · n = 6**
**pm = (-0.09, 4.26) mas/yr**




**Noise · n = 23**

